

2023 Consumer Confidence Report

For

Sawyer Hill Ecovillage

Berlin, Massachusetts

PWS ID #2028017

This report is a snapshot of the drinking water quality provided last year and is required of all water systems. It is updated and distributed annually and summarizes all the water quality testing done on your water supply in 2023, as well as the details about your water sources and distribution water system. Please take time to review this report and save it as a reference.

PUBLIC WATER SYSTEM INFORMATION

Address: Village Court, Berlin, MA 01503

Contact Person: Ethan Vaill Bickford

Telephone #: 617-794-4510

Email: water@sawyerhill.org

Website: <https://sawyerhill.org/>

Water System Improvements

Our water system is routinely inspected by the Massachusetts Department of Environmental Protection (MassDEP), who inspects this system for its technical, financial, and managerial capacity to provide safe drinking water. Your water system is operated by *Small Water Systems Services, LLC (SWSS)*, a contract drinking water and wastewater operations company, who oversees the routine operations of our system. As part of our ongoing commitment to you, last year we made the following improvements to our system:

- Replaced levels transducers in storage tanks to ensure level settings are activate appropriate actions
- Well hatches resealed

Opportunities for Public Participation

If you would like to participate in discussions regarding your water quality, you may attend the following meetings or educational events. Please contact Ethan Bickford at 617-794-4510 or email water@sawyerhill.org for more information about upcoming meetings.

YOUR DRINKING WATER SOURCE

Where Does My Drinking Water Come From?

Your water is provided by the following sources listed below:

Source Name	MassDEP Source ID#	Source Type	Location of Source
Well #1	2028017-01G	Groundwater	Northeast of the Ecovillage in the woods
Well #2	2028017-02G	Groundwater	Northeast of the Ecovillage in the woods

Is My Water Treated?

Our water system makes every effort to provide you with safe and pure drinking water. To improve the quality of the water delivered to you, we treat it to remove several contaminants. We have an emergency disinfection system installed to protect you against microbial contaminants.

The water quality of our system is constantly monitored by us and MassDEP to determine the effectiveness of existing water treatment and to determine if any additional treatment is required.

How Are These Sources Protected?

In 2001, MassDEP prepared Source Water Assessment Program (SWAP) Reports for water supply sources serving consumers at that time. The SWAP Report assesses the environmental susceptibility of public drinking water sources. Since Sawyer Hill Ecovillage was built after that date, no SWAP report is yet available. *SWSS* routinely monitors your wellhead for potential sources of contamination.

What Can Be Done to Improve Protection?

Residents can help protect sources by:

- Supporting water supply protection initiatives at the next HOA meeting.
- Taking hazardous household chemicals to hazardous material collection days.
- Limiting pesticide and fertilizer use, etc.

SUBSTANCES FOUND IN TAP WATER

Sources of drinking water (both tap water and bottled water) include rivers, lakes, streams, ponds, reservoirs, springs, and wells. As water travels over the surface of the land or through the ground, it dissolves naturally occurring minerals, and in some cases, radioactive material, and can pick up substances resulting from the presence of animals or from human activity.

Contaminants that may be present in source water include:

Microbial contaminants -such as viruses and bacteria, which may come from sewage treatment plants, septic systems, agricultural livestock operations, and wildlife.

Inorganic contaminants -such as salts and metals, which can be naturally occurring or result from urban stormwater runoff, industrial, or domestic wastewater discharges, oil and gas production, mining, and farming.

Pesticides and herbicides -which may come from a variety of sources such as agriculture, urban stormwater runoff, and residential uses.

Organic chemical contaminants -including synthetic and volatile organic chemicals, which are by-products of industrial processes and petroleum production, and can also come from gas stations, urban stormwater runoff, and septic systems.

Radioactive contaminants -which can be naturally occurring or be the result of oil and gas production and mining activities.

In order to ensure that tap water is safe to drink, the Department of Environmental Protection (MassDEP) and U.S. Environmental Protection Agency (EPA) prescribe regulations that limit the amount of certain contaminants in water provided by public water systems. The Food and Drug Administration (FDA) and Massachusetts Department of Public Health (DPH) regulations establish limits for contaminants in bottled water that must provide the same protection for public health.

All drinking water, including bottled water, may reasonably be expected to contain at least small amounts of some contaminants. The presence of contaminants does not necessarily indicate that water poses a health risk. More information about contaminants and potential health effects can be obtained by calling the EPA's Safe Drinking Water Hotline (800-426-4791).

Some people may be more vulnerable to contaminants in drinking water than the general population. Immuno-compromised persons such as persons with cancer undergoing chemotherapy, persons who have undergone organ transplants, people with HIV/AIDS or other immune system disorders, some elderly, and some infants can be particularly at risk from infections. These people should seek advice about drinking water from their health care providers. EPA/Centers for Disease Control and Prevention (CDC) guidelines on lowering the risk of infection by cryptosporidium and other microbial contaminants are available from the Safe Drinking Water Hotline (800-426-4791).

If present, elevated levels of lead can cause serious health problems, especially for pregnant women and young children. Lead in drinking water is primarily from materials and components associated with service lines and home plumbing. Sawyer Hill Ecovillage is responsible for providing high quality drinking water but cannot control the variety of materials used in plumbing components. When your water has been sitting for several hours, you can minimize the potential for lead exposure by flushing your tap for 30 seconds to 2 minutes before using water for drinking or cooking. If you are concerned about lead in your water, you may wish to have your water tested. Information on lead in drinking water, testing methods, and steps you can take to minimize exposure is available from the Safe Drinking Water Hotline or at <http://www.epa.gov/safewater/lead>.

IMPORTANT DEFINITIONS

Maximum Contaminant Level (MCL) – The highest level of a contaminant that is allowed in drinking water. MCLs are set as close to the MCLGs as feasible using the best available treatment technology.

Maximum Contaminant Level Goal (MCLG) –The level of a contaminant in drinking water below which there is no known or expected risk to health. MCLGs allow for a margin of safety.

Action Level (AL) – The concentration of a contaminant which, if exceeded, triggers treatment or other requirements that a water system must follow.

90th Percentile – Out of every 10 homes sampled, 9 were at or below this level.

Secondary Maximum Contaminant Level (SMCL) – These standards are developed to protect the aesthetic qualities of drinking water and are not health based.

Unregulated Contaminants

Unregulated contaminants are those for which EPA has not established drinking water standards. The purpose of unregulated monitoring is to assist EPA in determining their occurrence in drinking water and whether future regulation is warranted.

Massachusetts Office of Research and Standards Guideline (ORSG) – This is the concentration of a chemical in drinking water, at or below which, adverse health effects are unlikely to occur after chronic (lifetime) exposure. If exceeded, it serves as an indicator of the potential need for further action.

Treatment Technique (TT) – A required process intended to reduce the level of a contaminant in drinking water.

Running Annual Average (RAA) – The average of four consecutive quarters of data.

Maximum Residual Disinfectant Level (MRDL) -- The highest level of a disinfectant (chlorine, chloramines, chlorine dioxide) allowed in drinking water. There is convincing evidence that addition of a disinfectant is necessary for control of microbial contaminants.

Maximum Residual Disinfectant Level Goal (MRDLG) -- The level of a drinking water disinfectant (chlorine, chloramines, chlorine dioxide) below which there is no known expected risk to health.

MRDLG's do not reflect the benefits of the use of disinfectants to control microbial contaminants.

Total Trihalomethanes (TTHM) – A byproduct of drinking water chlorination.

Level 2 Assessment - A Level 2 assessment is a very detailed study of the water system to identify potential problems and determine (if possible) why an E. coli MCL violation has occurred and/or why total coliform bacteria have been found in our water system on multiple occasions.

ppm = parts per million, or milligrams per liter (mg/l)
 ppb = parts per billion, or micrograms per liter (ug/l)
 ppt = parts per trillion, or nanograms per liter
 pCi/l = picocuries per liter (a measure of radioactivity)
 NTU = Nephelometric Turbidity Units

ND = Not Detected
 N/A = Not Applicable
 mrem/year = millirems per year (a measure of radiation absorbed by the body)

WATER QUALITY TESTING RESULTS

What Does This Data Represent?

The water quality information presented in the following tables is from the most recent round of testing done in accordance with the regulations. All data shown was collected during the last calendar year unless otherwise noted.

MassDEP has reduced the monitoring requirements for Synthetic Organic Contaminants (SOC), Inorganic Contaminants (IOC), and Perchlorate for both wells, and Volatile Organic Contaminants for Well #2, because the sources are not at risk of contamination for these contaminants. VOCs are still required to be monitored in well #1 due to a history of detections. The last sample collected for IOCs, SOCs and VOCs on well #2 2/16/2021 and were found to meet all applicable US EPA and MassDEP standards.

Lead & Copper	Date(s) Collected	90 TH percentile	Action Level	MCLG	# of sites sampled	# of sites above Action Level	Possible Source of Contamination
Lead (ppb)	8/27/21 – 9/1/21	3	15	0	10	0	Corrosion of household plumbing systems; Erosion of natural deposits
Copper (ppm)	8/27/21 – 9/1/21	0.814	1.3	1.3	10	0	Corrosion of household plumbing systems; Erosion of natural deposits; Leaching from wood preservatives

Bacteria	MCL / TT	MCLG	Value	Date	Violation (Y/N)	Possible Sources
Total Coliform Bacteria	MCL	0	Positive	6/26/23, 6/27/23	Y	Human and animal fecal waste
Coliforms are bacteria that are naturally present in the environment and are used as an indicator that other, potentially harmful, waterborne pathogens may be present or that a potential pathway exists through which contamination may enter the drinking water distribution system. We found coliforms indicating the need to look for potential problems in water treatment or distribution. When this occurs, we are required to conduct assessments to identify any problems that were found during these assessments. During the past year one Level 2 Assessment was required to be completed for our water system. One Level 2 Assessment was completed. In addition, we were required to take corrective actions and all were completed.						

Regulated Contaminant	Date(s) Collected	Highest Result or Highest Running Average Detected	Range Detected	MCL or MRDL	MCLG or MRDLG	Violation (Y/N)	Possible Source(s) of Contamination
Inorganic Contaminants							
Barium (ppm)	2/16/21	0.009	0.007 – 0.009	2	2	N	Discharge of drilling wastes; discharge from metal refineries; erosion of natural deposits
Nitrate (ppm)	2/1/23	0.11	--	10	10	N	Runoff from fertilizer use; leaching from septic tanks; sewage; erosion of natural deposits
Volatile Organic Contaminants							
Toluene (ppm)	2/1/23	ND	--	1	1	N	Leaks and spills from gasoline and petroleum storage tanks; discharge from petroleum factories
Radioactive Contaminants							
Gross Alpha (pCi/l) (minus uranium)	2/16/21	6.6 (Well #1)	N/A	15	0	N	Erosion of natural deposits
Disinfectants and Disinfectant By-Products							
Chlorine (ppm) (free, total or combined)	1/16/23- 2/8/23, 7/7/23 – 11/28/23	0.30	0.15 – 0.41	4	4	N	Water additive used to control microbes
TTHM	2/1/23	3.9	--	80	N/A	N	By-product of drinking water disinfection

Unregulated and Secondary Contaminants

Unregulated contaminants are those for which there are no established drinking water standards. The purpose of unregulated contaminant monitoring is to assist regulatory agencies in determining their occurrence in drinking water and whether future regulation is warranted.

Unregulated Contaminants	Date(s) Collected	Result or Range Detected	Average Detected	SMCL	ORSG	Possible Source
Nickel (ppb)	2/16/21	ND - 3	1.5	--	100	Discharge from domestic wastewater, landfills, and mining and smelting operations
Bromodichloromethane (ppb)	2/1/23	1.4	--	--	--	Trihalomethane; by-product of drinking water chlorination

Chloroform (ppb)	2/1/23	1.2	--	--	70	Trihalomethane; by-product of drinking water chlorination [In non-chlorinated sources, chloroform may be naturally occurring.]
Dibromochloromethane	2/1/23	1.3	--	--	--	Trihalomethane; by-product of drinking water chlorination
Sodium (ppm)	2/1/23	11.6	--	--	20	Discharge from the use and improper storage of sodium-containing de-icing compounds or in water-softening agents
Secondary Contaminants						
Aluminum (ppb)	2/1/23	5	--	--	200	Residue from water treatment process; erosion of natural deposits.
Chloride (ppm)	2/1/23	18.3	--	--	250	Runoff and leaching from natural deposits; seawater influence
Iron (ppb)	2/1/23	47	--	300	N/A	Naturally occurring, corrosion of cast iron pipes
Manganese* (ppb)	2/1/23	ND	--	50	Health Advisory of 300	Natural sources as well as discharges from industrial uses
* EPA has established a lifetime Health Advisory (HA) for manganese of 0.3 mg/L and an acute HA at 1.0 mg/L						
Odor (T.O.N.)	2/1/23	2	--	3	N/A	Erosion of natural deposits; Leaching from wood preservatives.
Ph	2/1/23	6.6	--	6.5-8.5	N/A	Runoff and leaching from natural deposits; seawater influence
Sulfate (ppm)	2/1/23	71.4	--	250	N/A	Runoff and leaching from natural deposits; industrial wastes
Total Dissolved Solids (TDS) (ppm)	2/1/23	214	--	500	N/A	Erosion of natural deposits.
Zinc (ppm)	2/1/23	0.350	--	5	N/A	Erosion of natural deposits, leaching from plumbing materials

COMPLIANCE WITH DRINKING WATER REGS

Does My Drinking Water Meet Current Health Standards?

We are committed to providing you with the best water quality available. However, some contaminants that were tested last year did not meet all applicable health standards regulated by the state and federal government. Due to contaminant violations of Total Coliform during the periods of June and November 2022, our system took the following corrective actions.

During the Level 1 and Level 2 Assessments, corrective actions were taken to assess the cleanliness and structural integrity of the storage tanks and access hatches. During and after the work, the water in the storage tank was disinfected to eliminate coliform bacteria.

Our water system and MassDEP monitor and record the effectiveness of actions taken in response to contaminant violations. The health effect statement for total coliform is listed below.

HEALTH EFFECTS STATEMENTS

Lead: Infants and children who drink water containing lead in excess of the action level could experience delays in their physical or mental development. Children could show slight deficits in attention span and learning abilities. Adults who drink this water over many years could develop kidney problems or high blood pressure.

Nickel: Some people who drink water containing nickel at high concentrations for many years could experience effects on the lung, stomach, blood, liver, kidneys, immune system, reproduction, and development.

Sodium: Sodium sensitive individuals, such as those experiencing hypertension, kidney failure, or congestive heart failure, should be aware of the sodium levels where exposures are being carefully controlled.

Total Coliform: Coliforms are bacteria that are naturally present in the environment and are used as an indicator that other, potentially harmful, waterborne pathogens may be present or that a potential pathway exists through which contamination may enter the drinking water distribution system. We found coliforms indicating the need to look for potential problems in water treatment or distribution. When this occurs, we are required to conduct assessments to identify any problems that were found during these assessments.

EDUCATIONAL INFORMATION

Cross-Connection Control and Backflow Prevention

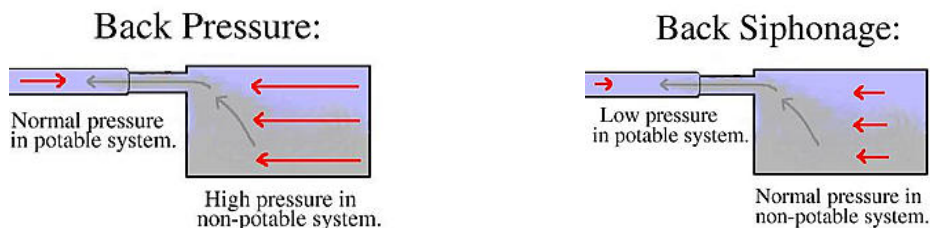
Sawyer Hill Ecovillage makes every effort to ensure that the water delivered to your home and business is clean, safe and free of contamination. Our staff works very hard to protect the quality of the water delivered to our customers from the time the water is extracted via deep wells from underground aquifers or withdrawal point from a surface water source, throughout the entire treatment and distribution system. But what happens when the water reaches your home or business? Is there still a need to protect the water quality from contamination caused by a cross-connection? If so, how?

What is a cross-connection?

A cross-connection occurs whenever the drinking water supply is or could be in contact with potential sources of pollution or contamination. Cross-connections exist in piping arrangements or equipment that allows the drinking water to come in contact with non-potable liquids, solids, or gases (hazardous to humans) in event of a backflow.

What is a backflow?

Backflow is the undesired reverse of the water flow in the drinking water distribution lines. This backward flow of water can occur when the pressure created by equipment or a system such as a boiler or air-conditioning is higher than the water pressure inside the water distribution line (back pressure), or when the pressure in the distribution line drops due to routine occurrences such as water main breaks or heavy water demand causing the water to flow backward inside the water distribution system (back siphonage). Backflow is a problem that many water consumers are unaware of, a problem that each and every water customer has a responsibility to help prevent.



What can I do to help prevent a cross-connection?

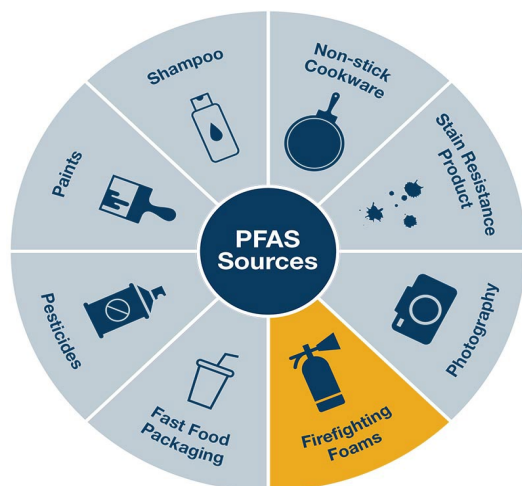
Without the proper protection something as simple as a garden hose has the potential to contaminate or pollute the drinking water lines in your house. In fact, over half of the country's cross-connection incidents involve unprotected garden hoses. There are very simple steps that you as a drinking water user can take to prevent such hazards, they are:

- NEVER submerge a hose in soapy water buckets, pet watering containers, pool, tubs, sinks, drains, or chemicals.
- NEVER attach a hose to a garden sprayer without the proper backflow preventer.
- Buy and install a hose bibb vacuum breaker in any threaded water fixture. The installation can be as easy as attaching a garden hose to a spigot. This inexpensive device is available at most hardware stores and home-improvement centers.
- Identify and be aware of potential cross-connections to your water line.
- Buy appliances and equipment with backflow preventers.
- Buy and install backflow prevention devices or assemblies for all high and moderate hazard connections.

If you are the owner or manager of a property that is being used as a commercial, industrial, or institutional facility you must have your property's plumbing system surveyed for cross-connection by your water purveyor. If your property has NOT been surveyed for cross-connection, contact your water department to schedule a cross-connection survey.

WHAT ARE PFAS AND WHY ARE THEY A PROBLEM?

Per- and polyfluoroalkyl substances (PFAS) are a family of chemicals used since the 1950s to manufacture stain-resistant, water-resistant, and non-stick products. PFAS are widely used in common consumer products such as coatings, on food packaging, outdoor clothing, carpets, leather goods, ski and snowboard waxes, and more. Certain types of firefighting foam—historically used by the U.S. military, local fire departments, and airports to fight oil and gasoline fires—may contain PFAS.



PFAS in drinking water is an important emerging issue nationwide. Because PFAS are water soluble, over time PFAS from some firefighting foam, manufacturing sites, landfills, spills, air deposition from factories and other releases can seep into surface soils. From there, PFAS can leach into groundwater or surface water, and can contaminate drinking water. PFAS have also been found in rivers, lakes, fish, and wildlife.

Exposure can occur when someone uses certain products that contain PFAS, eats PFAS-contaminated food, or drinks PFAS-contaminated water. When ingested, some PFAS can build up in the body and, over time, these PFAS may increase to a level where health effects could occur.

On October 2, 2020, MassDEP published its PFAS public drinking water standard, of 20 nanograms per liter (ng/L) (or ppt) – individually or for the sum of the concentrations of six specific PFAS. These PFAS are perfluorooctane sulfonic acid (PFOS); perfluorooctanoic acid

(PFOA); perfluorohexane sulfonic acid (PFHxS); perfluorononanoic acid (PFNA); perfluoroheptanoic acid (PFHpA); and perfluorodecanoic acid (PFDA). MassDEP abbreviates this set of six PFAS as “PFAS6.” This drinking water standard is set to be protective against adverse health effects for all people consuming the water.

For more information, see the MassDEP webpage regarding PFAS at: www.mass.gov/info-details/per-and-polyfluoroalkyl-substances-pfas, or the EPA webpage at: www.epa.gov/pfas/pfas-explained.

SWSS is working with MassDEP to ensure that all our community water systems have been and will be tested for PFAS. The results of that testing can be found in this and future Annual Water Quality Reports for your system.

ADDITIONAL INFORMATION

SWSS has been contracted on an annual basis to provide licensed water operator coverage for the water system serving Sawyer Hill Ecovillage. It is our responsibility to maintain the system’s compliance with all drinking water operation requirements. We monitor your drinking water, routinely evaluate the water quality entering your distribution system and inspect the systems regularly. For more information, call your operators at SWSS at 978-486-1008.